

06 — MEMORIZE THIS (facts you must know cold)

A. NUMBERED LISTS (with mnemonics)

5 CMM levels — mnemonic: "I Really Don't Manage Overtime"

1. **I**nitial — chaotic, individual effort, not repeatable
2. **R**epeatable — basic project management
3. **D**efined — own standard process
4. **M**anaged — data collection & analysis
5. **O**ptimizing — continuous improvement + innovation

5 TMM levels — mnemonic: "I Prefer Increasingly Mature Orgs"

1. **I**nitial — testing after code, ad hoc, no tracking
2. **P**hase Definition — testing/debugging goals, initiate test planning
3. **I**ntegration — establish test group, integrate testing into lifecycle
4. **M**anagement and Measurement — org-wide review, test management, evaluate quality
5. **O**ptimization / Defect Prevention & QC — defect prevention, statistical QC, optimization

7 Principles of Testing — mnemonic: "Please Eat Every Delicious Pizza Carefully, Alright"

1. **P**resence of defects (not absence)
2. **E**xhaustive testing is impossible
3. **E**arly testing
4. **D**efect clustering
5. **P**esticide paradox
6. **C**ontext dependent
7. **A**bsence-of-errors fallacy

6 ISO-9126 characteristics — mnemonic: "Fat Rabbits Usually Eat More Peas"

Functionality · **R**eliability · **U**sability · **E**fficiency · **M**aintainability · **P**ortability (*Security is NOT one of them.*)

5 Views of quality (Kitchenham & Pfleeger) — mnemonic: "My Uncle Makes Pretty Vases"

Mystical · **U**ser · **M**anufacturing · **P**roduct · **V**alue-based

3 QA categories — mnemonic: "Prevent, Reduce, Contain" (PRC)

Prevention (before injection) · **R**eduction (remove faults) · **C**ontainment (limit failures)

4 Testing levels + who — mnemonic: "U-I-S-A: Devs Test Test Users"

Unit → Developer · **I**ntegration → Tester · **S**ystem → Tester · **A**cceptance → End-user

3 SqE stages — mnemonic: "PED"

Planning · **E**xecution · **D**ecision (*SQE generic testing process phases: Pre-QA → In-QA → Post-QA.*)

Other short lists to keep

- **2 ways of defect prevention:** error source removal · error blocking
- **2 ways of defect containment:** software fault-tolerance (NVP, recovery) · safety assurance
- **3 testing team models:** Vertical · Horizontal · Mixed
- **3 project challenges:** Time · Cost · Scope

- **3 reasons to improve a process:** Quality · Lead Time · Cost
- **4 TPI improvement steps:** determine area → evaluate current → identify next → implement
- **4 process variations:** Waterfall · Iterative · Spiral · Agile (+ Maintenance)

B. NAMED PEOPLE / MODELS

Name / Model	What it is / who
McCall	Quality Factors and Criteria model
Kitchenham & Pfleeger	The five views of software quality
Ilene Burnstein	Pioneered the TMM (Testing Maturity Model)
ISO-9126	Most influential quality framework — 6 characteristics
CMM	Capability Maturity Model — evaluates development process (5 levels)
TMM	Testing Maturity Model — evaluates testing process (5 levels)
TPI	Test Process Improvement — improves the testing process (4 steps)
ISO 9001	Standard based on the manufacturing view (with CMM)
Naik & Tripathy / Jeff Tian	The two course textbook authors

C. FAMOUS INCIDENTS

Incident	What happened
Therac-25	Radiation-therapy machine by AECL (1986) ; a software bug caused massive radiation overdoses and deaths.
Ariane 5	On 4 June 1996 , the rocket tore itself apart 37 seconds after launch due to a control-software malfunction — the most expensive computer bug in history .

D. EVERY "HOW MANY" FACT

Question	Answer
CMM levels	5
TMM levels	5
Principles of testing	7
ISO-9126 top-level characteristics	6
Views of quality (Kitchenham & Pfleeger)	5
QA categories	3
Testing levels	4
SqE stages	3
Generic testing-process phases	3 (Pre/In/Post-QA)
Ways to do defect prevention	2
Ways to do defect containment	2
Testing team models	3
TPI improvement steps	4
Software types (Buy/Build)	2
Project challenges	3
Ariane 5 seconds to failure	37

E. ONE-LINE DEFINITIONS TO NAIL

- **Error** = human action producing an incorrect result.
- **Fault** = incorrect step/process/data definition in a program.
- **Failure** = inability of a system to perform its required function.
- **Defect** = behavioural deviation from spec (umbrella term = "bug").
- **Reliability** = probability of executing without failure for a specific period of time.
- **Safety** = accident-free. **Accident** = failure with severe consequences. **Hazard** = pre-condition to an accident.
- **Test plan** = high-level; **Test case** = low-level; **Test suite** = collection run until stopping criteria.
- **Regression testing** = reuse existing tests (design none) to check nothing broke.
- **White-box** = structural/implementation; **Black-box** = functional/specification; **Gray-box** = mixed.
- **SQA** = prevent, no execution, verification; **SQC** = detect, executes, validation.